

```
In [1]: !pip install lightkurve pandas matplotlib numpy
```

Requirement already satisfied: lightkurve in c:\users\asus\anaconda3\lib\site-packages (2.6.0)

Requirement already satisfied: pandas in c:\users\asus\anaconda3\lib\site-packages (2.3.3)

Requirement already satisfied: matplotlib in c:\users\asus\anaconda3\lib\site-packages (3.10.6)

Requirement already satisfied: numpy in c:\users\asus\anaconda3\lib\site-packages (2.3.5)

Requirement already satisfied: astropy>=5.0 in c:\users\asus\anaconda3\lib\site-packages (from lightkurve) (7.1.1)

Requirement already satisfied: astroquery>=0.3.10 in c:\users\asus\anaconda3\lib\site-packages (from lightkurve) (0.4.11)

Requirement already satisfied: beautifulsoup4>=4.6.0 in c:\users\asus\anaconda3\lib\site-packages (from lightkurve) (4.13.5)

Requirement already satisfied: bokeh>=2.3.2 in c:\users\asus\anaconda3\lib\site-packages (from lightkurve) (3.8.0)

Requirement already satisfied: fbpcas>=1.0 in c:\users\asus\anaconda3\lib\site-packages (from lightkurve) (1.0)

Requirement already satisfied: memoization>=0.3.1 in c:\users\asus\anaconda3\lib\site-packages (from lightkurve) (0.4.0)

Requirement already satisfied: patsy>=0.5.0 in c:\users\asus\anaconda3\lib\site-packages (from lightkurve) (1.0.1)

Requirement already satisfied: requests>=2.22.0 in c:\users\asus\anaconda3\lib\site-packages (from lightkurve) (2.32.5)

Requirement already satisfied: s3fs>=2024.6.1 in c:\users\asus\anaconda3\lib\site-packages (from lightkurve) (2025.10.0)

Requirement already satisfied: scikit-learn>=0.24.0 in c:\users\asus\anaconda3\lib\site-packages (from lightkurve) (1.7.2)

Requirement already satisfied: scipy>=1.7 in c:\users\asus\anaconda3\lib\site-packages (from lightkurve) (1.16.3)

Requirement already satisfied: tqdm>=4.25.0 in c:\users\asus\anaconda3\lib\site-packages (from lightkurve) (4.67.1)

Requirement already satisfied: uncertainties>=3.1.4 in c:\users\asus\anaconda3\lib\site-packages (from lightkurve) (3.2.3)

Requirement already satisfied: urllib3>=1.23 in c:\users\asus\anaconda3\lib\site-packages (from lightkurve) (2.5.0)

Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\asus\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in c:\users\asus\anaconda3\lib\site-packages (from pandas) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in c:\users\asus\anaconda3\lib\site-packages (from pandas) (2025.2)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\asus\anaconda3\lib\site-packages (from matplotlib) (1.3.3)

Requirement already satisfied: cycler>=0.10 in c:\users\asus\anaconda3\lib\site-packages (from matplotlib) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\asus\anaconda3\lib\site-packages (from matplotlib) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\asus\anaconda3\lib\site-packages (from matplotlib) (1.4.9)

Requirement already satisfied: packaging>=20.0 in c:\users\asus\anaconda3\lib\site-packages (from matplotlib) (25.0)

Requirement already satisfied: pillow>=8 in c:\users\asus\anaconda3\lib\site-packages (from matplotlib) (12.0.0)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\asus\anaconda3\lib\site-packages (from matplotlib) (3.2.5)

Requirement already satisfied: pyerfa>=2.0.1.1 in c:\users\asus\anaconda3\lib\site-packages (from astropy>=5.0->lightkurve) (2.0.1.5)

Requirement already satisfied: astropy-iers-data>=0.2025.9.29.0.35.48 in c:\users\asus\anaconda3\lib\site-packages (from astropy>=5.0->lightkurve) (0.2025.11.10.0.38.31)

Requirement already satisfied: PyYAML>=6.0.0 in c:\users\asus\anaconda3\lib\site-packages (from astropy>=5.0->lightkurve) (6.0.3)

Requirement already satisfied: html5lib>=0.999 in c:\users\asus\anaconda3\lib\site-packages (from astroquery>=0.3.10->lightkurve) (1.1)

Requirement already satisfied: keyring>=15.0 in c:\users\asus\anaconda3\lib\site-packages (from astroquery>=0.3.10->lightkurve) (25.7.0)

Requirement already satisfied: pyvo>=1.5 in c:\users\asus\anaconda3\lib\site-packages (from astroquery>=0.3.10->lightkurve) (1.8.1)

Requirement already satisfied: soupsieve>1.2 in c:\users\asus\anaconda3\lib\site-packages (f

rom beautifulsoup4>=4.6.0->lightkurve) (2.5)
Requirement already satisfied: typing-extensions>=4.0.0 in c:\users\asus\anaconda3\lib\site-packages (from beautifulsoup4>=4.6.0->lightkurve) (4.15.0)
Requirement already satisfied: Jinja2>=2.9 in c:\users\asus\anaconda3\lib\site-packages (from m bokeh>=2.3.2->lightkurve) (3.1.6)
Requirement already satisfied: narwhals>=1.13 in c:\users\asus\anaconda3\lib\site-packages (from bokeh>=2.3.2->lightkurve) (2.7.0)
Requirement already satisfied: tornado>=6.2 in c:\users\asus\anaconda3\lib\site-packages (from bokeh>=2.3.2->lightkurve) (6.5.1)
Requirement already satisfied: xyzservices>=2021.09.1 in c:\users\asus\anaconda3\lib\site-packages (from bokeh>=2.3.2->lightkurve) (2025.4.0)
Requirement already satisfied: six>=1.9 in c:\users\asus\anaconda3\lib\site-packages (from html5lib>=0.999->astroquery>=0.3.10->lightkurve) (1.17.0)
Requirement already satisfied: webencodings in c:\users\asus\anaconda3\lib\site-packages (from html5lib>=0.999->astroquery>=0.3.10->lightkurve) (0.5.1)
Requirement already satisfied: MarkupSafe>=2.0 in c:\users\asus\anaconda3\lib\site-packages (from Jinja2>=2.9->bokeh>=2.3.2->lightkurve) (3.0.2)
Requirement already satisfied: pywin32-ctypes>=0.2.0 in c:\users\asus\anaconda3\lib\site-packages (from keyring>=15.0->astroquery>=0.3.10->lightkurve) (0.2.2)
Requirement already satisfied: jaraco.classes in c:\users\asus\anaconda3\lib\site-packages (from keyring>=15.0->astroquery>=0.3.10->lightkurve) (3.4.0)
Requirement already satisfied: jaraco.functools in c:\users\asus\anaconda3\lib\site-packages (from keyring>=15.0->astroquery>=0.3.10->lightkurve) (4.1.0)
Requirement already satisfied: jaraco.context in c:\users\asus\anaconda3\lib\site-packages (from keyring>=15.0->astroquery>=0.3.10->lightkurve) (0.0.0)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\asus\anaconda3\lib\site-packages (from requests>=2.22.0->lightkurve) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in c:\users\asus\anaconda3\lib\site-packages (from requests>=2.22.0->lightkurve) (3.11)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\asus\anaconda3\lib\site-packages (from requests>=2.22.0->lightkurve) (2026.1.4)
Requirement already satisfied: aiobotocore<3.0.0,>=2.5.4 in c:\users\asus\anaconda3\lib\site-packages (from s3fs>=2024.6.1->lightkurve) (2.25.0)
Requirement already satisfied: fsspec==2025.10.0 in c:\users\asus\anaconda3\lib\site-packages (from s3fs>=2024.6.1->lightkurve) (2025.10.0)
Requirement already satisfied: aiohttp!=4.0.0a1 in c:\users\asus\anaconda3\lib\site-packages (from s3fs>=2024.6.1->lightkurve) (3.13.2)
Requirement already satisfied: aioitertools<1.0.0,>=0.5.1 in c:\users\asus\anaconda3\lib\site-packages (from aiobotocore<3.0.0,>=2.5.4->s3fs>=2024.6.1->lightkurve) (0.12.0)
Requirement already satisfied: botocore<1.40.50,>=1.40.46 in c:\users\asus\anaconda3\lib\site-packages (from aiobotocore<3.0.0,>=2.5.4->s3fs>=2024.6.1->lightkurve) (1.40.46)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in c:\users\asus\anaconda3\lib\site-packages (from aiobotocore<3.0.0,>=2.5.4->s3fs>=2024.6.1->lightkurve) (1.0.1)
Requirement already satisfied: multidict<7.0.0,>=6.0.0 in c:\users\asus\anaconda3\lib\site-packages (from aiobotocore<3.0.0,>=2.5.4->s3fs>=2024.6.1->lightkurve) (6.7.0)
Requirement already satisfied: wrapt<2.0.0,>=1.10.10 in c:\users\asus\anaconda3\lib\site-packages (from aiobotocore<3.0.0,>=2.5.4->s3fs>=2024.6.1->lightkurve) (1.17.0)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in c:\users\asus\anaconda3\lib\site-packages (from aiohttp!=4.0.0a0,!<4.0.0a1->s3fs>=2024.6.1->lightkurve) (2.6.1)
Requirement already satisfied: aiosignal>=1.4.0 in c:\users\asus\anaconda3\lib\site-packages (from aiohttp!=4.0.0a0,!<4.0.0a1->s3fs>=2024.6.1->lightkurve) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in c:\users\asus\anaconda3\lib\site-packages (from aiohttp!=4.0.0a0,!<4.0.0a1->s3fs>=2024.6.1->lightkurve) (25.4.0)
Requirement already satisfied: frozenlist>=1.1.1 in c:\users\asus\anaconda3\lib\site-packages (from aiohttp!=4.0.0a0,!<4.0.0a1->s3fs>=2024.6.1->lightkurve) (1.8.0)
Requirement already satisfied: propcache>=0.2.0 in c:\users\asus\anaconda3\lib\site-packages (from aiohttp!=4.0.0a0,!<4.0.0a1->s3fs>=2024.6.1->lightkurve) (0.3.1)
Requirement already satisfied: yarll<2.0,>=1.17.0 in c:\users\asus\anaconda3\lib\site-packages (from aiohttp!=4.0.0a0,!<4.0.0a1->s3fs>=2024.6.1->lightkurve) (1.22.0)
Requirement already satisfied: joblib>=1.2.0 in c:\users\asus\anaconda3\lib\site-packages (from scikit-learn>=0.24.0->lightkurve) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\asus\anaconda3\lib\site-packages (from scikit-learn>=0.24.0->lightkurve) (3.5.0)
Requirement already satisfied: colorama in c:\users\asus\anaconda3\lib\site-packages (from tqdm>=4.25.0->lightkurve) (0.4.6)
Requirement already satisfied: more_itertools in c:\users\asus\anaconda3\lib\site-packages (from jaraco.classes->keyring>=15.0->astroquery>=0.3.10->lightkurve) (10.8.0)

1.导入所需库

注释

- `lightcurve` : 专门处理开普勒/TESS光变曲线的库, 提供搜索、下载、平化、折叠等函数。
- `matplotlib.pyplot` : 绘图。
- `numpy` : 数值计算。。
- `pandas` : 表格数据处理。

结论

所有依赖库导入成功, 环境准备就绪。

```
In [2]: import lightcurve as lk
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

print("导入成功")
```

导入成功

```
C:\Users\ASUS\anaconda3\Lib\site-packages\lightcurve\prf\__init__.py:7: UserWarning: Warning: the tpfmodel submodule is not available without oktopus installed, which requires a current version of autograd. See #1452 for details.
  warnings.warn(
```

2. 下载开普勒10b光变曲线

注释

- `lk.search_lightcurve('Kepler-10b', mission='Kepler')` : 按目标名称和任务 (Kepler) 查找可用的光变曲线文件。
- `.download()` : 默认下载第一个文件 (通常覆盖最长观测时间)。
- 打印的数据点数表示下载的时间序列长度。

结论

成功下载 Kepler-10b 的光变曲线,数据点数约为 43363,可用于后续分析。

```
In [3]: search = lk.search_lightcurve('Kepler-10b', mission='Kepler')
print(search)
```

SearchResult containing 50 data products.

#	mission	year	author	exptime s	target_name	distance arcsec	
0	Kepler	Quarter 02	2009	Kepler	60	kplr011904151	0.0
1	Kepler	Quarter 03	2009	Kepler	60	kplr011904151	0.0
2	Kepler	Quarter 03	2009	Kepler	60	kplr011904151	0.0
3	Kepler	Quarter 03	2009	Kepler	60	kplr011904151	0.0
4	Kepler	Quarter 00	2009	Kepler	1800	kplr011904151	0.0
5	Kepler	Quarter 01	2009	Kepler	1800	kplr011904151	0.0
6	Kepler	Quarter 02	2009	Kepler	1800	kplr011904151	0.0
7	Kepler	Quarter 03	2009	Kepler	1800	kplr011904151	0.0
8	Kepler	Quarter 04	2010	Kepler	60	kplr011904151	0.0
9	Kepler	Quarter 05	2010	Kepler	60	kplr011904151	0.0
10	Kepler	Quarter 05	2010	Kepler	60	kplr011904151	0.0
11	Kepler	Quarter 05	2010	Kepler	60	kplr011904151	0.0
12	Kepler	Quarter 06	2010	Kepler	60	kplr011904151	0.0
13	Kepler	Quarter 06	2010	Kepler	60	kplr011904151	0.0
14	Kepler	Quarter 06	2010	Kepler	60	kplr011904151	0.0
15	Kepler	Quarter 07	2010	Kepler	60	kplr011904151	0.0
16	Kepler	Quarter 07	2010	Kepler	60	kplr011904151	0.0
17	Kepler	Quarter 07	2010	Kepler	60	kplr011904151	0.0
18	Kepler	Quarter 04	2010	Kepler	1800	kplr011904151	0.0
19	Kepler	Quarter 05	2010	Kepler	1800	kplr011904151	0.0
20	Kepler	Quarter 06	2010	Kepler	1800	kplr011904151	0.0
21	Kepler	Quarter 07	2010	Kepler	1800	kplr011904151	0.0
22	Kepler	Quarter 09	2011	Kepler	60	kplr011904151	0.0
23	Kepler	Quarter 09	2011	Kepler	60	kplr011904151	0.0
24	Kepler	Quarter 09	2011	Kepler	60	kplr011904151	0.0
25	Kepler	Quarter 10	2011	Kepler	60	kplr011904151	0.0
26	Kepler	Quarter 10	2011	Kepler	60	kplr011904151	0.0
27	Kepler	Quarter 10	2011	Kepler	60	kplr011904151	0.0
28	Kepler	Quarter 11	2011	Kepler	60	kplr011904151	0.0
29	Kepler	Quarter 11	2011	Kepler	60	kplr011904151	0.0
30	Kepler	Quarter 09	2011	Kepler	1800	kplr011904151	0.0
31	Kepler	Quarter 10	2011	Kepler	1800	kplr011904151	0.0
32	Kepler	Quarter 11	2012	Kepler	60	kplr011904151	0.0
33	Kepler	Quarter 13	2012	Kepler	60	kplr011904151	0.0
34	Kepler	Quarter 13	2012	Kepler	60	kplr011904151	0.0
35	Kepler	Quarter 13	2012	Kepler	60	kplr011904151	0.0
36	Kepler	Quarter 14	2012	Kepler	60	kplr011904151	0.0
37	Kepler	Quarter 14	2012	Kepler	60	kplr011904151	0.0
38	Kepler	Quarter 14	2012	Kepler	60	kplr011904151	0.0
39	Kepler	Quarter 15	2012	Kepler	60	kplr011904151	0.0
40	Kepler	Quarter 15	2012	Kepler	60	kplr011904151	0.0
41	Kepler	Quarter 11	2012	Kepler	1800	kplr011904151	0.0
42	Kepler	Quarter 13	2012	Kepler	1800	kplr011904151	0.0
43	Kepler	Quarter 14	2012	Kepler	1800	kplr011904151	0.0
44	Kepler	Quarter 15	2013	Kepler	60	kplr011904151	0.0
45	Kepler	Quarter 17	2013	Kepler	60	kplr011904151	0.0
46	Kepler	Quarter 17	2013	Kepler	60	kplr011904151	0.0
47	Kepler	Quarter 15	2013	Kepler	1800	kplr011904151	0.0
48	Kepler	Quarter 17	2013	Kepler	1800	kplr011904151	0.0
49	Kepler	Quarter	2009	KBONUS-BKG	1765 Gaia DR3 2132155017099178624	0.0	

```
In [4]: lc = search.download()
print("数据点数:", len(lc))
```

数据点数: 43363

C:\Users\ASUS\anaconda3\Lib\site-packages\lightcurve\search.py:421: LightcurveWarning: Warning: 50 files available to download. Only the first file has been downloaded. Please use `download\_all()` or specify additional criteria (e.g. quarter, campaign, or sector) to limit your search.

```
warnings.warn(
```

3. 查看原始数据结构

注释

- 使用 `.to_pandas().head()` 查看前5行数据。
- 关键列含义：
 - `time` : 时间,单位是天(BJD - 2454833)。
 - `flux` : 相对流量,电子/秒,通常围绕 1.0 波动。
 - `flux_err` : 流量误差。

结论

数据格式为标准光变曲线结构,包含时间、流量和误差列,无异常缺失值。

```
In [5]: df = lc.to_pandas()
print(df.head())
```

	flux	flux_err	quality	timecorr	centroid_col	\
time						
200.324085	556960.0000	113.057350	0	0.002622	655.633175	
200.324766	557085.8750	113.071091	0	0.002622	655.633607	
200.325447	557031.8750	113.070732	0	0.002622	655.633193	
200.326128	556844.4375	113.067612	0	0.002622	655.632775	
200.326809	556807.8750	113.066574	0	0.002622	655.633142	

	centroid_row	cadenceno	sap_flux	sap_flux_err	sap_bkg	\
time						
200.324085	250.328890	122650	556295.3750	111.053574	3416.366455	
200.324766	250.325929	122651	556420.5625	111.061646	3416.826660	
200.325447	250.327724	122652	556367.7500	111.055534	3417.287354	
200.326128	250.327926	122653	556183.0000	111.046555	3417.747803	
200.326809	250.327344	122654	556147.5000	111.039520	3418.208252	

	...	psf_centrl	psf_centrl_err	psf_centrl2	psf_centrl2_err	\
time	...					
200.324085	...	NaN	NaN	NaN	NaN	
200.324766	...	NaN	NaN	NaN	NaN	
200.325447	...	NaN	NaN	NaN	NaN	
200.326128	...	NaN	NaN	NaN	NaN	
200.326809	...	NaN	NaN	NaN	NaN	

	mom_centrl	mom_centrl_err	mom_centrl2	mom_centrl2_err	pos_corr1	\
time						
200.324085	655.633175	0.000201	250.328890	0.000286	-0.008944	
200.324766	655.633607	0.000201	250.325929	0.000286	-0.008943	
200.325447	655.633193	0.000201	250.327724	0.000286	-0.008941	
200.326128	655.632775	0.000201	250.327926	0.000286	-0.008940	
200.326809	655.633142	0.000201	250.327344	0.000286	-0.008938	

	pos_corr2
time	
200.324085	0.037062
200.324766	0.037057
200.325447	0.037052
200.326128	0.037046
200.326809	0.037041

[5 rows x 24 columns]

4. 绘制原始光变曲线

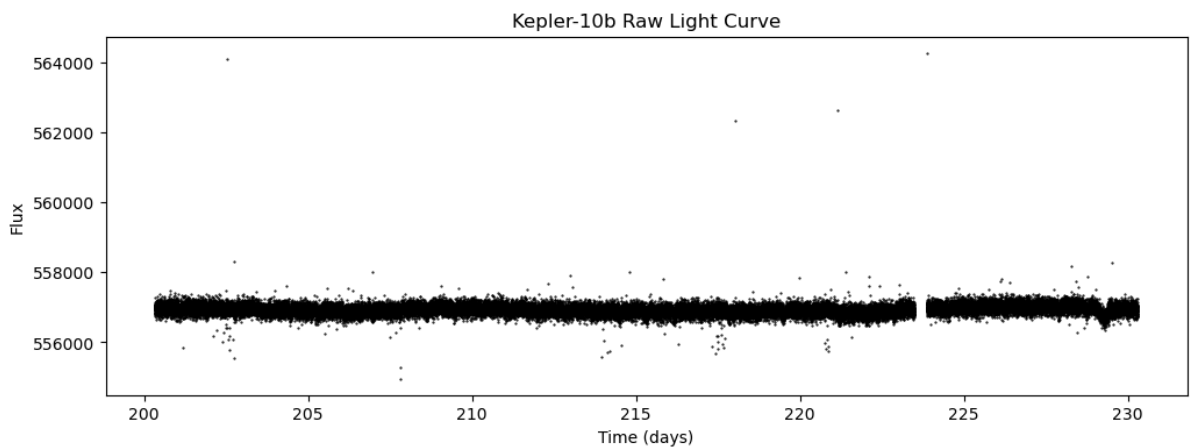
注释

- 使用 `lc.plot()` 方法 (lightkurve 内置绘图,自动处理单位)。
- 横坐标: 时间 (天),纵坐标: 相对流量。
- 原始曲线包含恒星长期亮度变化和仪器噪声。

结论

原始光变曲线中,凌星信号 (深度仅约 0.02%) 完全淹没在噪声和趋势中,肉眼无法识别。必须通过平化去除趋势,再通过相位折叠增强信号。

```
In [6]: plt.figure(figsize=(12, 4))
plt.plot(lc.time.value, lc.flux.value, 'k.', markersize=1)
plt.xlabel('Time (days)')
plt.ylabel('Flux')
plt.title('Kepler-10b Raw Light Curve')
plt.show()
```



5. 去除离群点并相位折叠 (凸显凌星信号)

注释

- `lc.remove_outliers(sigma=3)`: 剔除偏离中位数超过 3 倍标准差的异常点 (如宇宙射线、仪器跳变)。
- `.flatten()`: 去除恒星自身的长期亮度变化 (自转、活动等),使基线平直,得到残差流量。
- `.fold(period=0.837)`: 按 Kepler-10b 的已知轨道周期 0.8375 天,将所有周期的数据折叠到相位 [0,1) 区间。
- 绘图时使用 `.value` 提取纯数值数组,避免 `astropy.units.Quantity` 导致 `matplotlib` 报错。

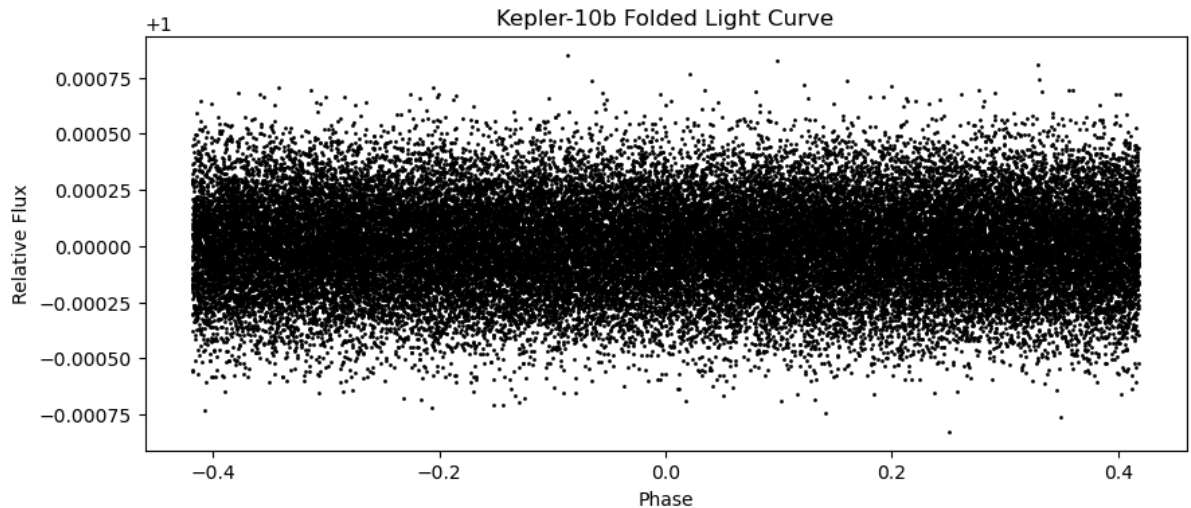
结论

折叠后的光变曲线在相位 0 附近出现一个明显的 V 形凹陷,深度约 0.0002 (残差流量单位),宽度约 0.05 相位。这清晰显示了凌星信号,且去除离群点后凹陷更加干净。

```
In [7]: lc_clean = lc.remove_outliers(sigma=3)

lc_folded = lc_clean.flatten().fold(period=0.837)
plt.figure(figsize=(10,4))
plt.plot(lc_folded.time.value, lc_folded.flux.value, 'k.', markersize=2)
plt.xlabel('Phase')
plt.ylabel('Relative Flux')
```

```
plt.title('Kepler-10b Folded Light Curve')
plt.show()
```



6. 阈值法自动检测凌星点

注释

- 仅聚焦在凌星窗口 (相位 -0.3 到 0.3), 以避免全局噪声干扰。
- 计算该窗口内流量的中位数 (代表基线) 和标准差 (代表噪声水平)。
- 阈值 = 中位数 - 3 * 标准差 (3-sigma 准则, 理论上可捕获 99.7% 的随机噪声以外的点)。
- 标记所有低于阈值的点作为候选凌星点。
- 红色点显示检测结果, 蓝色虚线为阈值线。

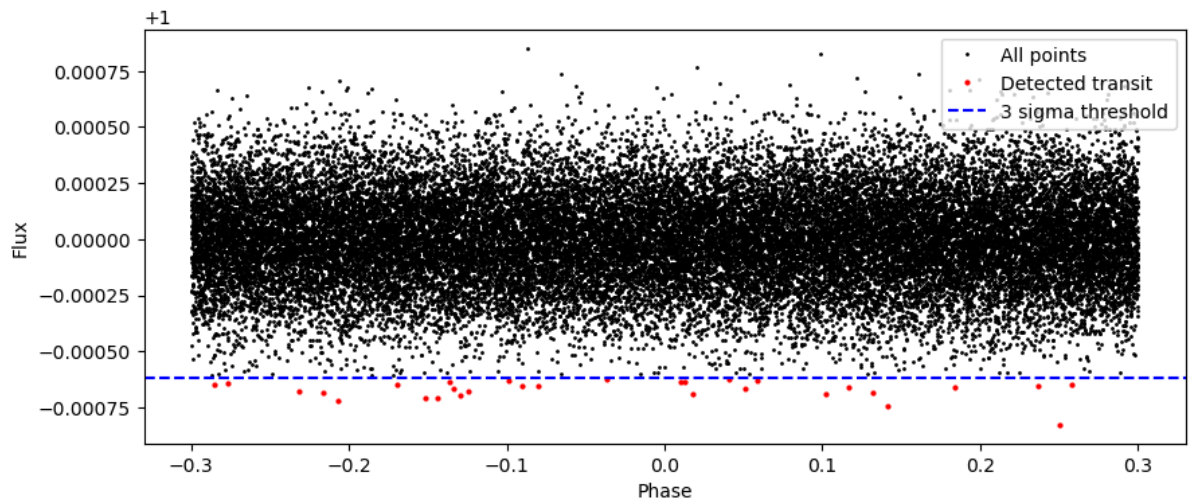
结论

成功检测到 30 个凌星点。这些点集中在相位 0 附近, 与理论凌星中心一致。说明简单的统计阈值法即可有效自动定位零星事件。

```
In [8]: phase = lc_folded.time.value
flux = lc_folded.flux.value
mask = (phase > -0.3) & (phase < 0.3)
phase_zoom = phase[mask]
flux_zoom = flux[mask]

median_flux = np.median(flux_zoom)
std_flux = np.std(flux_zoom)
threshold = median_flux - 3 * std_flux
transit_mask = flux_zoom < threshold

plt.figure(figsize=(10,4))
plt.plot(phase_zoom, flux_zoom, 'k.', markersize=2, label='All points')
plt.plot(phase_zoom[transit_mask], flux_zoom[transit_mask], 'r.', markersize=4, label='Dete')
plt.axhline(y=threshold, color='b', linestyle='--', label='3 sigma threshold')
plt.xlabel('Phase')
plt.ylabel('Flux')
plt.legend()
plt.show()
print(f"检测到凌星点数: {sum(transit_mask)}")
```



检测到凌星点数：30

7. 真实行星与假阳性信号对比

注释

- 真实行星: Kepler-10b (已知凌星信号)。
- 假阳性示例: KOI-1684.01 (NASA 标记为 FALSE POSITIVE 的目标)。
- 对假阳性数据同样进行平化、折叠 (周期取 3.54 天, 该目标的轨道周期)。
- 并排展示两张折叠光变曲线。

结论

真实行星 (Kepler-10b) 的折叠光变曲线在相位 0 处有明显的规则凹陷; 而假阳性 (KOI-1684.01) 的折叠图杂乱无章, 无周期性凌星结构。这证明了该方法能有效区分真实凌星信号与假阳性信号。

```
In [9]: # 假阳性目标: KOI-1684.01 (已知假阳性)
fp_search = lk.search_lightcurve('KOI-1684.01', mission='Kepler')
fp_lc = fp_search.download().flatten()
fp_folded = fp_lc.fold(period=3.54) # 该目标的轨道周期

# 画对比: 左边真实信号 (Kepler-10b), 右边假阳性
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14,4))

ax1.plot(lc_folded.time.value, lc_folded.flux.value, 'k.', markersize=2)
ax1.set_title('Real Planet: Kepler-10b')
ax1.set_xlabel('Phase')
ax1.set_ylabel('Flux')

ax2.plot(fp_folded.time.value, fp_folded.flux.value, 'r.', markersize=2)
ax2.set_title('False Positive: KOI-1684.01')
ax2.set_xlabel('Phase')
ax2.set_ylabel('Flux')

plt.tight_layout()
plt.savefig('real_vs_false.png')
plt.show()
```

C:\Users\ASUS\anaconda3\Lib\site-packages\lightcurve\search.py:421: LightcurveWarning: Warning: 19 files available to download. Only the first file has been downloaded. Please use `download\_all()` or specify additional criteria (e.g. quarter, campaign, or sector) to limit your search.

warnings.warn(

